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V2. Arithmetic Null Models

A pre-registered battery of statistics against five families of null models

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A pre-registered battery of statistics against five families of null models

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Problem statement

GUE statistics occur widely in systems unrelated to the zeta function: universality is an attractor, not evidence. Before attributing arithmetic specificity to AB-cloud, one must measure what the mere chaos of the lattice produces. Null models destroy the arithmetic while keeping the geometry and scales.

The key discipline is pre-registration: the statistic vector $S = (\langle r \rangle, \Sigma^2(L), \Delta_3(L), K(t), D_{KS})$ was fixed before the runs, no metric is chosen post hoc, and the multiple-testing correction is Bonferroni over five families. This is a direct implementation of Solution 2 of the plan.

Method

Families: M0 — protocol gates ($N_v \in \{0,2,8,16,32\}$, $q = 1$, $W = 0.5$, $\alpha = 1/2$); N1 — all link phases replaced by uniform $U(-\pi, \pi]$; N3 — geometry without vortices (Hofstadter plus disorder); N4 — random vortex positions outside the protocol; plus reference ensembles GUE, GOE and Poisson of the same size 576. The full vector S is measured on a 200-level unfolded window for each model.

The D_{KS} reference is ensemble-based: the pooled $\{r_n\}$ sample of sixteen GUE matrices (3168 values) serves as the empirical GUE reference instead of a spacing surmise — this removes small-window systematics.

Table 1. Model ladder ($\langle r \rangle$ and D_{KS} to GUE)

Model	$\langle r \rangle$	$D_{KS} \to$ GUE	Class
M0 AB-cloud	0.6093	0.0499	GUE
N1 random phases	0.5837	0.0925	GUE
N3 no vortices	0.2733	0.5518	intermediate
N4 random positions	0.6181	0.0492	GUE
reference GUE	0.5984	0.0622	GUE
reference GOE	0.5256	0.1622	GOE
reference Poisson	0.4198	0.3150	Poisson

Results and interpretation

AB-cloud falls into the GUE class together with N1 and N4: the vortex mechanics is sufficient for chaos but not specific — exactly what universality predicts. N3 (no vortices) stays intermediate: half flux alone does not fully chaotise a finite lattice. The reference ladder Poisson → GOE → GUE is reproduced, validating the battery itself.

Hence the methodological constraint of the series: $\langle r \rangle$ and D_{KS} carry no arithmetic information — specificity must be sought only in arithmetic-sensitive trace-level metrics (V7) and in the inverse problem (V10). This feature split is the central methodological output of the plan, and V2 delivers it.

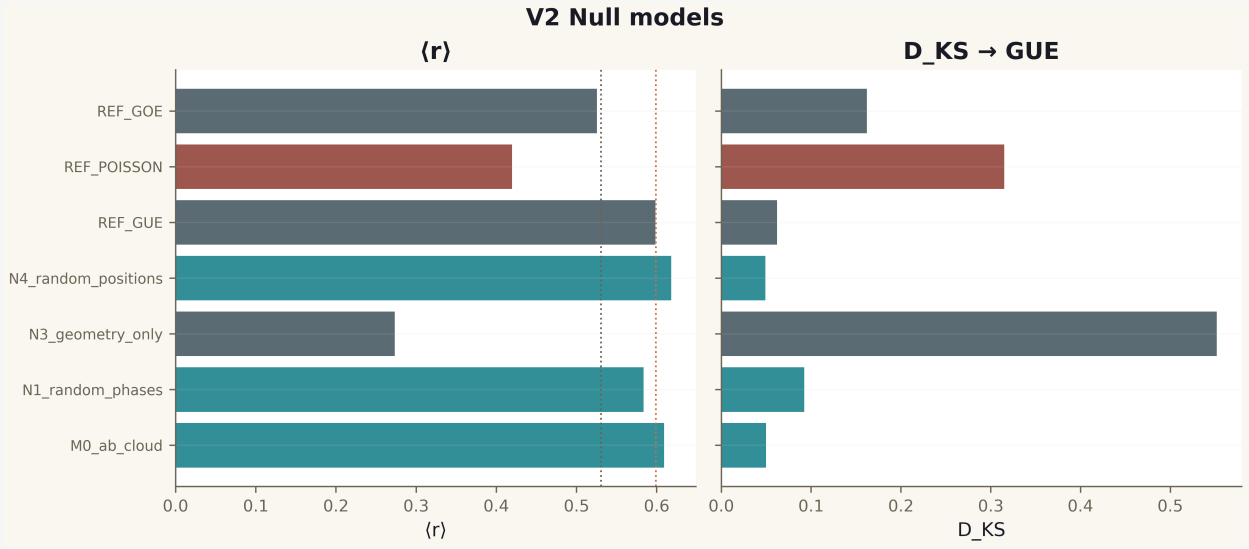


Fig. 1. Campaign V2 data (600 dpi, file figures_hires).

The pre-registered battery is built; the class of AB-cloud is GUE, as for any sufficiently chaotic vortex lattice. Any further claim about arithmetic must pass through trace metrics.